



OrcaRouter

ORCAROUTER SECURITY RESEARCH

The AI Threat Report 2026

How enterprises are attacked through their AI systems — and the control stack that stops it

ABOUT THIS REPORT

The AI Threat Report is the annual flagship study of OrcaRouter Security Research, the security research team behind the OrcaRouter AI gateway. The 2026 edition maps how enterprises are attacked through their AI systems — drawing on the public incident record, analyst and government data, and the threat patterns documented in OrcaRouter's security research — and sets out the control architecture that contains them.

All statistics are drawn from named public sources, cited as endnotes. Exhibits labeled *illustrative* encode analytical judgment rather than measured data; the Methodology section explains how to read each. Nothing in this report is derived from OrcaRouter customer data.

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The year in eight numbers

Eight numbers that define the AI security year — drawn from the public record and the research compiled in this report.

88%

of organizations now use AI in at least one business function — adoption is no longer the frontier ¹

3%

of organizations breached through AI had proper AI access controls in place; the other 97% did not ²

32%

of security leaders saw prompt-based attacks on their AI applications in the past twelve months ³

42 sec

average time for a successful attack on a production LLM application, start to finish ⁴

62%

of organizations experienced a deepfake attack in the past year ³

+\$670K

the additional breach cost when “shadow AI” is involved — \$4.63M vs. a \$4.44M global average ⁵

\$893M

in losses across 22,364 AI-related complaints in the FBI's first-ever AI crime accounting ⁶

Aug 2

2026 — the EU AI Act becomes fully applicable; AI governance moves from posture to legal obligation ⁷

Five findings for 2026

In 2025, enterprise AI crossed a line: systems that drafted text became systems that take actions. The threat landscape crossed the same line, on the same schedule — and in most organizations, the controls did not. This report maps how enterprises are actually attacked through their AI systems, and lays out the control architecture that contains it.

01 Adoption outran governance — and attackers noticed first.

Eighty-eight percent of organizations now run AI in at least one business function¹, but only 37% assess the security of AI tools before deploying them⁸. The result is measurable: 13% of organizations have already had an AI model or application breached, and 97% of those lacked basic AI access controls². Ungoverned “shadow AI” now adds an average \$670,000 premium to breach costs⁵.

02 The attack surface moved into the model's context window.

The defining vulnerability of the era is prompt injection — ranked LLM01 in the OWASP Top 10 for LLM Applications⁹. It is not a bug that will be patched; it is a structural property of models that cannot reliably separate instructions from data. Every document, web page, email, and tool result an AI system reads is now part of your attack surface. The objective is containment, not elimination.

03 2025 was the year attacks became agentic.

Zero-click exfiltration through enterprise copilots (EchoLeak¹⁰), cross-repository data theft through agent tooling (GitHub MCP¹¹), service-side leaks in autonomous research agents (ShadowLeak¹²), and the first reported AI-orchestrated espionage campaign¹³ moved AI risk from embarrassing text to real-world actions: data exfiltrated, credentials abused, infrastructure manipulated — increasingly with no human in the loop.

04 The economics now favor the attacker.

Successful attacks on production LLM applications take 42 seconds on average⁴; 90% of them leak sensitive data⁴. Deepfake attempts occur at industrial frequency¹⁴ and 62% of organizations have already faced one³. Meanwhile, agents introduce a loss class that needs no breach at all: denial-of-wallet, where a hijacked or runaway agent simply spends. Defense must be cheap, automatic, and in the request path to compete.

05 The defense that works is architectural.

Organizations that contain AI threats do four things: give every agent a scoped, expiring identity; screen content on the way in and out; police actions — tool calls, network egress, spend — at a policy gateway; and write every decision to an audit trail they can hand to a regulator. This is zero trust applied to AI, it maps cleanly onto OWASP, NIST AI RMF, ISO/IEC 42001, and the EU AI Act⁷, and it can be deployed at the gateway without rewriting a line of agent code.

1

THE LANDSCAPE

The state of AI security in 2026

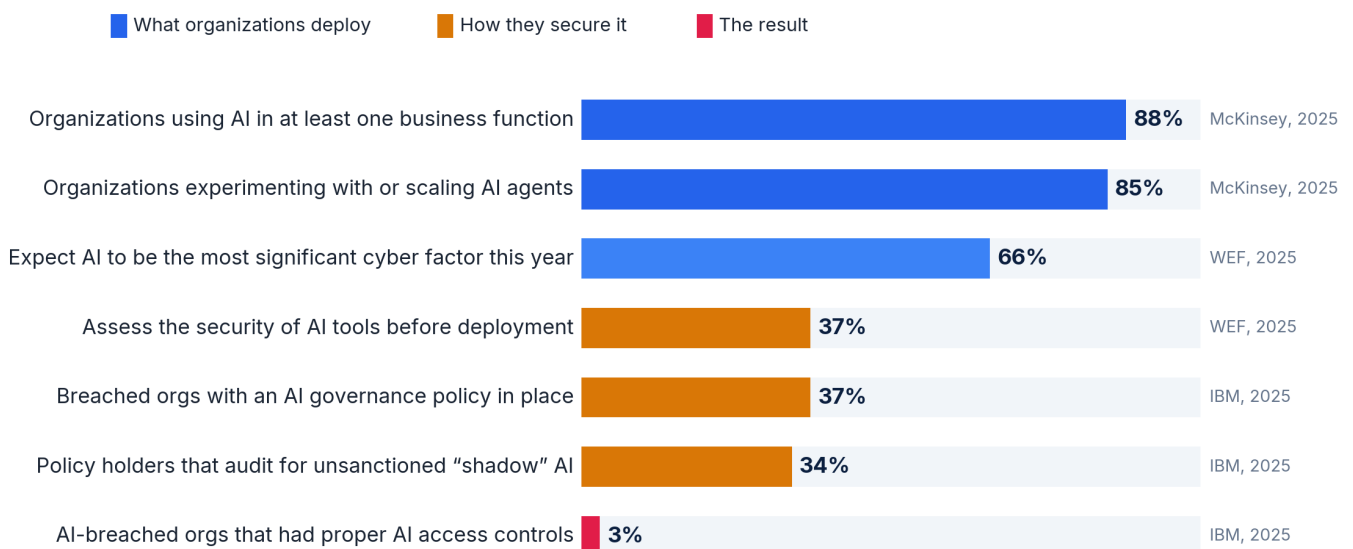
Enterprise AI went from pilot to production in eighteen months. Its security model did not. The gap between what organizations deploy and what they control is now the most reliably exploited seam in the enterprise.

1.1 Adoption outran governance

The adoption race is over: 88% of organizations report using AI in at least one business function, up from 78% a year earlier — and 62% are already experimenting with AI agents, with another 23% scaling them¹. Organizational AI use jumped from 55% in 2023 to 78% in 2024¹⁵ and has kept climbing. AI is no longer a pilot program; it is infrastructure.

Governance has not kept pace, and the gap is quantifiable. Two-thirds of organizations expect AI to be the most significant cybersecurity factor of the coming year — yet only 37% have a process to assess the security of an AI tool *before* it is deployed⁸. Among organizations that suffered a breach, 63% either have no AI governance policy or are still writing one; of those that do have a policy, only 34% audit for unsanctioned AI use⁵.

EXHIBIT 2 Adoption has outrun readiness — and the outcome is showing up in breach data



Source: McKinsey, The State of AI in 2025; World Economic Forum, Global Cybersecurity Outlook 2025; IBM, Cost of a Data Breach Report 2025

The consequences stopped being hypothetical in 2025. Thirteen percent of organizations reported breaches of AI models or applications — and of those, 97% lacked proper AI access controls². Sixty percent of these AI incidents ended in compromised data; 31% disrupted operations². One in five organizations traced a breach to *shadow AI* — tools employees adopted without security's knowledge — and those breaches ran \$670,000 above the global average cost, \$4.63 million against \$4.44 million⁵.

Read those numbers together and the diagnosis is uncomfortable but clear: enterprises did not deploy AI insecurely because the controls failed. They deployed it before the controls existed.

1.2 Three years from curiosity to kill chain

The public incident record tells the story of the threat's maturation in three acts. **2023 was the text era:** the harms were leaks and embarrassments — source code pasted into chatbots¹⁶, a caching bug exposing other users' conversations¹⁷ — and the seminal research on indirect prompt injection¹⁸ read as a warning about a future problem. **2024 was the liability era:** a tribunal held an airline responsible for its chatbot's invented refund policy¹⁹, a deepfaked video call extracted US\$25.6 million from a single engineering firm²⁰, and researchers demonstrated the first self-replicating worm targeting AI-powered applications²¹.

2025 was the agentic era. The incidents that defined the year shared a signature: no human clicked anything. A crafted email exfiltrated data from Microsoft 365 Copilot with zero user interaction (EchoLeak, CVE-2025-32711)¹⁰. A poisoned GitHub issue steered an agent into leaking private repositories¹¹. A research agent was tricked into leaking Gmail data from the provider's own infrastructure, where no enterprise control could see it (ShadowLeak)¹². OAuth tokens stolen from an AI chat integration cascaded into data theft across hundreds of downstream SaaS tenants²². The first malicious MCP server was caught in the wild, quietly BCC'ing every email that passed through it²³. And in November, Anthropic disclosed what it assessed as the first cyber espionage campaign in which an AI agent executed the large majority of the operation — target reconnaissance, exploit development, credential harvesting — with humans intervening at a handful of decision points¹³ (a characterization, it should be noted, that independent researchers could not fully verify).

The first half of **2026** has confirmed the trajectory rather than bent it. An exposed database on a viral agent social network let anyone hijack any of 1.5 million registered agents²⁴; a coordinated campaign seeded an agent skill marketplace with hundreds of malware-bearing skills²⁵; and Microsoft published critical CVEs demonstrating that a single injected prompt can achieve host-level code execution in mainstream agent frameworks²⁶. The era's lesson compounds: every new layer of agent infrastructure — protocols, marketplaces, frameworks — arrives with the same trust gap the last one had.

EXHIBIT 1 Three years, three eras — from leaked text to autonomous operations



Source: Public disclosures and research as cited in the endnotes; OrcaRouter Security Research compilation

1.3 The perimeter moved into the context window

Traditional security architecture assumes a boundary: trusted inside, untrusted outside, controls at the seam. Language models dissolve that boundary, because a model's *input* is also its *programming*. Any text a model reads — a user's question, a retrieved document, a web page, a tool result — can carry instructions the model may follow. There is no reliable, general mechanism by which today's models distinguish content to *process* from commands to *obey*. That is why prompt injection holds the number-one position in the OWASP Top 10 for LLM Applications⁹, and why it cannot be fixed the way a buffer overflow is fixed.

Attackers have internalized this faster than defenders. In Gartner's 2025 survey, 32% of security leaders reported attacks on their AI applications through the prompt itself, and 29% reported attacks on GenAI application infrastructure³. Telemetry from production LLM applications shows successful attacks completing in 42 seconds on average, with 90% of them leaking sensitive data⁴. In HiddenLayer's survey, 74% of organizations said they definitely knew of an AI breach in 2024²⁷. And the AI-crime category now has an official price tag: the FBI's 2025 Internet Crime

Report — the first in its history with a dedicated AI section — logged 22,364 AI-related complaints and nearly \$893 million in losses⁶.

The same capability shift arms the other side of the ledger: 82.6% of phishing emails analyzed in late 2024 and early 2025 showed signs of AI generation, a figure that reached 86% by April 2026²⁸; a deepfake attempt occurred every five minutes during 2024¹⁴; and by late 2025, Google's threat intelligence team had catalogued the first malware families that call LLMs *mid-execution* to regenerate their own code and evade detection²⁹. AI is simultaneously the new attack surface and the new attack tool.

1.4 What changes in 2026

Three forces converge this year and convert AI security from initiative to obligation. **Regulation arrives:** the EU AI Act, in force since August 2024, becomes fully applicable on August 2, 2026, completing a staged rollout that began with prohibitions in February 2025 and general-purpose-AI obligations in August 2025⁷. **Agents go mainstream:** Gartner projects 40% of enterprise applications will ship with task-specific AI agents by the end of 2026, up from under 5% in 2025³⁰ — and simultaneously predicts that by 2028 a quarter of enterprise GenAI applications will suffer at least five security incidents per year³¹. **The failure mode is priced in:** Gartner expects over 40% of agentic AI projects to be canceled by the end of 2027, with inadequate risk controls among the leading causes³².

The strategic implication threads through the rest of this report: AI security in 2026 is not a model problem to be solved with better training. It is an architecture problem — identity, content control, action control, and evidence — and it is solvable with the discipline enterprises already apply to every other production system.

2

THE THREATS

A taxonomy of AI threats

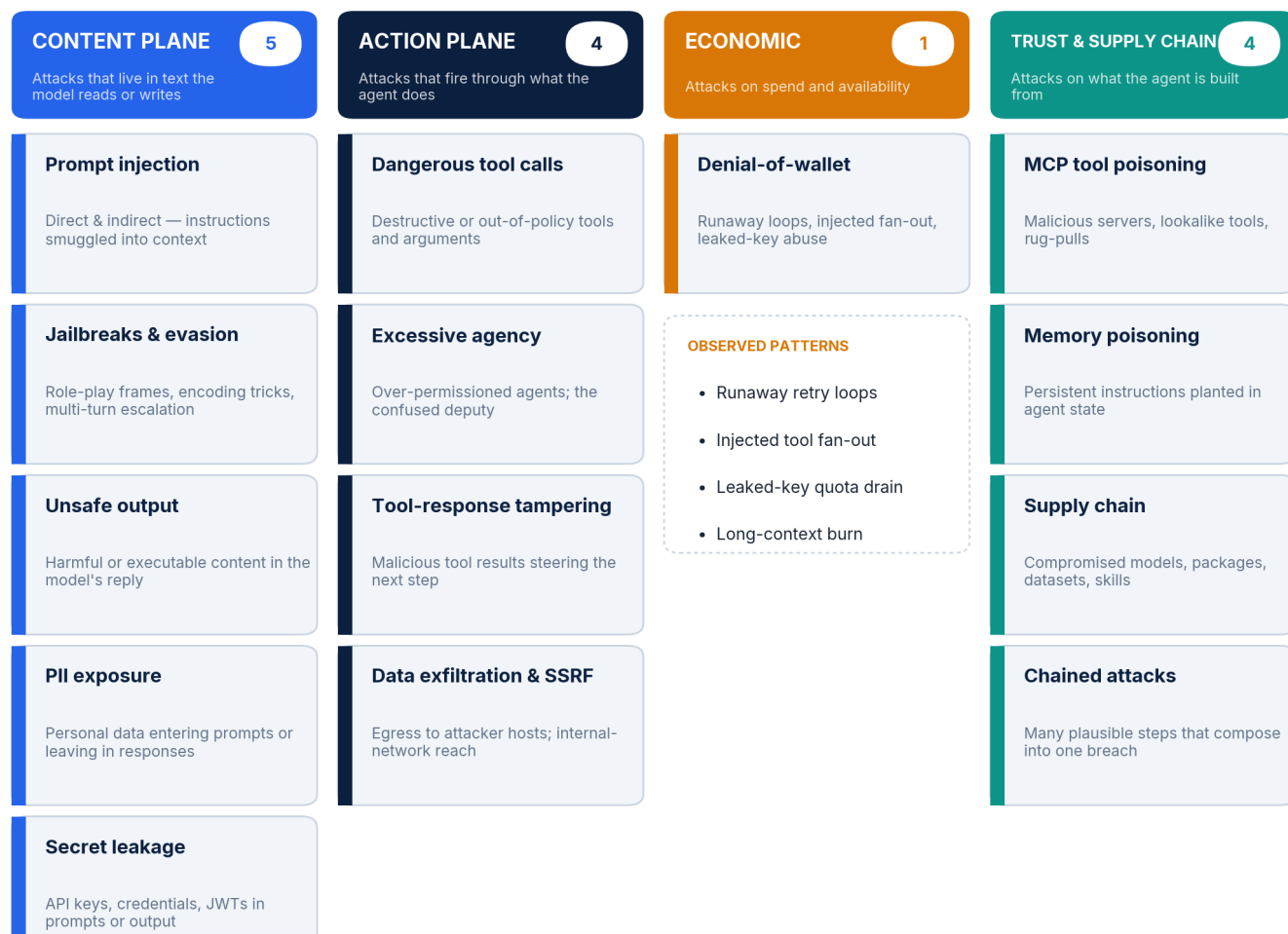
Fourteen threat classes account for the overwhelming majority of real-world AI attacks. They organize into four families — and, more usefully for defenders, onto two planes: what the model reads and writes, and what the agent does.

2.1 Four families, two planes

Security teams confronting AI for the first time tend to inherit a vocabulary invented piecemeal — jailbreaks, injections, hallucinations — that obscures more than it reveals. Our taxonomy, distilled from the attack patterns documented across OrcaRouter's threat research³³ and the public incident record, organizes the landscape into fourteen threat classes in four families: **content-plane** threats that live in the text a model reads or writes; **action-plane** threats that fire through what an agent *does*; **economic** threats that attack spend and availability; and **trust & supply chain** threats that compromise what the agent is built from.

The plane distinction is the load-bearing one, because the two planes are governed by different controls. Content-plane attacks are caught by screening text — input and output guardrails. Action-plane attacks are caught by policing behavior — rules over tool calls, network destinations, and spend. The most damaging real-world incidents chain both planes: an injection arrives as content, then cashes out as an action. A defense that watches only one plane will verifiably miss the attacks that matter.

EXHIBIT 3 Fourteen threat classes, four families — organized by where the attack lives



Source: OrcaRouter Security Research, from the OrcaRouter threat model³³

2.2 The content plane: attacks in the text

Prompt injection is the family's apex threat and the technique underlying most modern AI attacks. In its *direct* form, the attacker is the user: crafted input overrides the system prompt or unlocks restricted behavior. In its far more dangerous *indirect* form, the attacker plants instructions in content the AI will eventually ingest — a web page, a PDF, an email, a calendar invite, a ticket, a tool response. The model encounters the payload inside trusted context and executes it with the agent's full authority. No account compromise is required; the attacker never touches your system. They publish content and wait.

Jailbreaks are the adversarial craft of defeating a model's safety training: role-play frames, encoding tricks (Base64, character substitution, invisible Unicode tag bytes), multi-turn escalation, and token-stuffing. The technique pool is industrial — measured success rates of roughly 20% against guarded production applications⁴ — and evolves weekly, which is why static keyword filters age so badly. **Unsafe output** inverts the direction of risk: model responses carrying executable payloads (SQL fragments, scripts) into downstream systems that auto-execute them. **PII exposure** and **secret leakage** round out the family — personal data and live credentials flowing through prompts, logs, and completions, where they acquire new copies, new jurisdictions, and new readers. These two classes are as much compliance liabilities as attack vectors: they are how an AI deployment quietly becomes a GDPR, HIPAA, or PCI incident.

2.3 The action plane: attacks through behavior

Give a model tools and you have given every successful injection an execution engine. **Dangerous tool calls** — the model invoking destructive operations (`shell.exec`, `db.delete`, payment APIs) or legitimate tools with hostile arguments — are the action plane's bluntest threat. **Excessive agency** is its quietest: an agent holding more capability than its task requires, so that a single hijack inherits a large blast radius. The canonical failure is the *confused deputy* — an agent with legitimate authority tricked into exercising it for an attacker. Both EchoLeak and the GitHub MCP exploit were confused-deputy attacks: nothing was "hacked" except the agent's intentions¹⁰¹¹.

Tool-response tampering closes the loop from the other side: a compromised or malicious tool returns results that carry instructions or fabricated data, steering the agent's next step. And **data exfiltration over the network** is where most chains terminate — the agent fetches an attacker URL with stolen data encoded in the path, or reaches an internal service it should never see (the SSRF pattern). Egress is the choke point: an exfiltration that cannot leave is an incident report, not a breach.

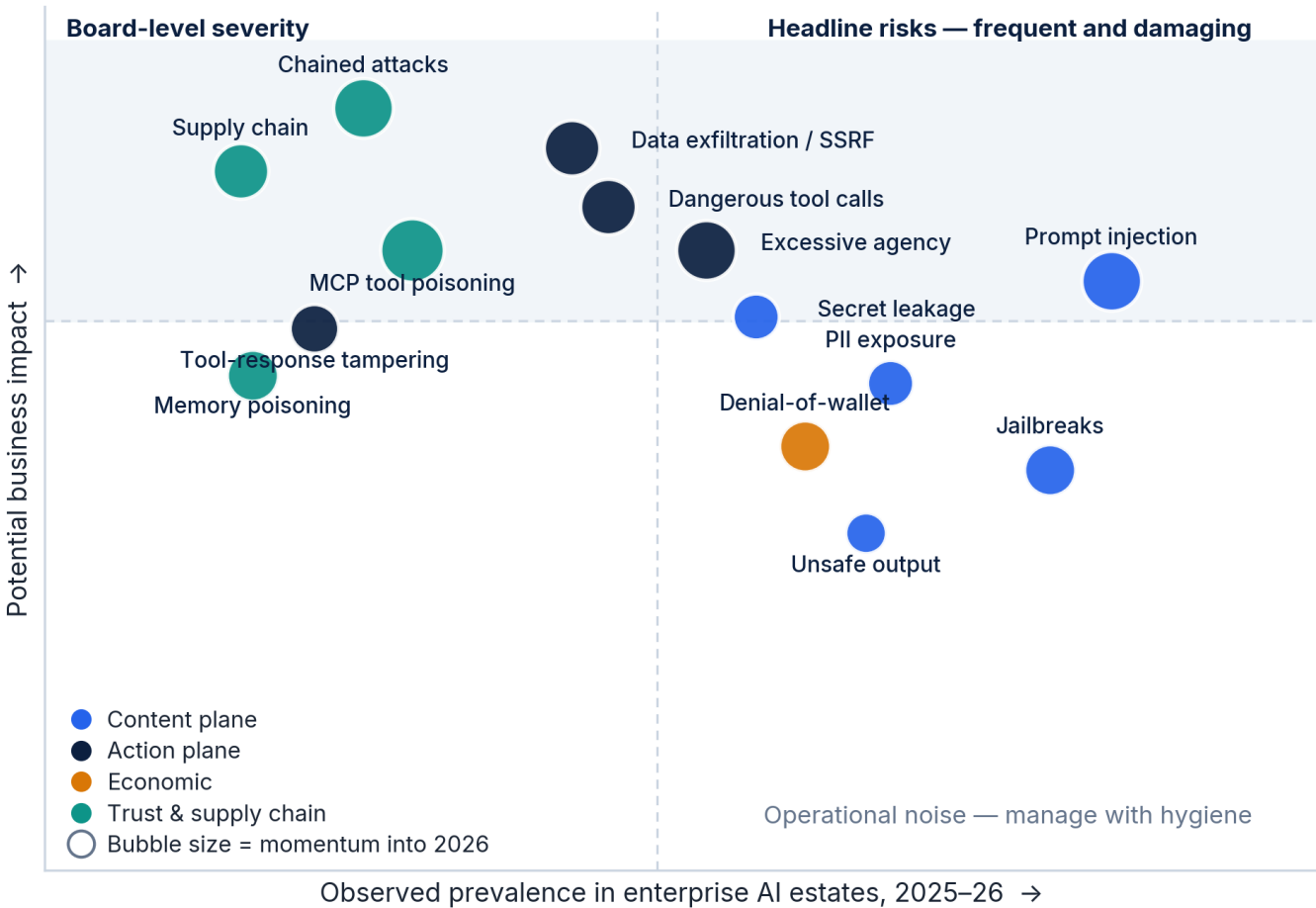
2.4 The economic family: attacks on the meter

Agents introduce a loss class that requires no data theft at all. **Denial-of-wallet** attacks the meter: a runaway retry loop, an injected instruction to fan out thousands of tool calls, a leaked key driven hard by a third party — the economy security researchers named "LLMjacking," with observed victim exposure topping \$46,000 a day³⁴ — or simply a long-context task burning tokens far beyond intent. The observed patterns are distinctive — the same call hammered in a tight window, spend spiking against an hourly baseline, fan-out the workspace has never exhibited — and they are *behavioral* signatures, invisible to any per-request check. Unbounded spend authority is the only vulnerability in this report that compounds by the minute.

2.5 Trust & supply chain: attacks on what the agent is made of

The fourth family compromises the agent's components rather than its conversation. **MCP tool poisoning** exploits the Model Context Protocol's trust model: an agent connects to a server and believes its self-declared tool manifest. A malicious server advertises legitimate-looking tools whose descriptions carry hidden instructions for the model — and the *rug-pull* variant behaves honestly for weeks before silently shipping a new manifest that adds dangerous capabilities. **Memory poisoning** plants instructions in an agent's persistent state, where they survive across sessions and re-execute long after the hostile content is gone. **Supply chain** covers the classic vector at new scale — compromised model weights (a single 2024 sweep found roughly one hundred backdoored models on a public model hub³⁵), poisoned datasets, malicious packages and skills, and weaponized configuration like the Rules File Backdoor attack on AI coding assistants³⁶. And **chained attacks** are the family's capstone: sequences of individually plausible steps whose damage lives entirely in the composition. Chapter 3 dissects them.

EXHIBIT 4 Severity and prevalence are inversely correlated — the rarest attacks are the ones that end up in board reports



Source: OrcaRouter Security Research analysis; illustrative; positions reflect judgment over the 2025-26 public record, not measured incidence

Exhibit 4 captures the landscape's central asymmetry. The high-frequency threats — injection attempts, jailbreak probes — are mostly absorbed by basic controls; the low-frequency ones — chained attacks, supply-chain compromise, network exfiltration — are the ones that produce eight-figure incidents. Defense budgets anchored to frequency alone will systematically underinvest in exactly the threats that matter.

2.6 Anchoring to the standards

The taxonomy maps directly onto the frameworks your auditors and regulators will reference. The OWASP Top 10 for LLM Applications 2025⁹ is the de-facto risk vocabulary; NIST's Generative AI Profile³⁷ supplies the risk-management spine; MITRE ATLAS catalogues adversary techniques. The table below carries the crosswalk; Chapter 5 extends it to controls and to the EU AI Act and ISO/IEC 42001.

Threat family	Representative classes	OWASP LLM Top 10 (2025)	Primary control plane
Content plane	Prompt injection · jailbreaks · unsafe output · PII · secrets	LLM01 Prompt Injection · LLM02 Sensitive Information Disclosure · LLM05 Improper Output Handling	Input & output guardrails
Action plane	Dangerous tool calls · excessive agency · tampering · exfiltration	LLM06 Excessive Agency · LLM05 Improper Output Handling	Action-layer firewall: tool, argument & egress policy
Economic	Denial-of-wallet and resource abuse	LLM10 Unbounded Consumption	Spend caps · per-run cost ceilings · anomaly detection
Trust & supply chain	MCP poisoning · memory poisoning · supply chain · chained attacks	LLM03 Supply Chain · LLM04 Data & Model Poisoning · LLM08 Vector & Embedding Weaknesses	Manifest pinning · quarantine · provenance · run correlation

Source: OWASP GenAI Security Project⁹; OrcaRouter Security Research mapping

3

THE ATTACKS

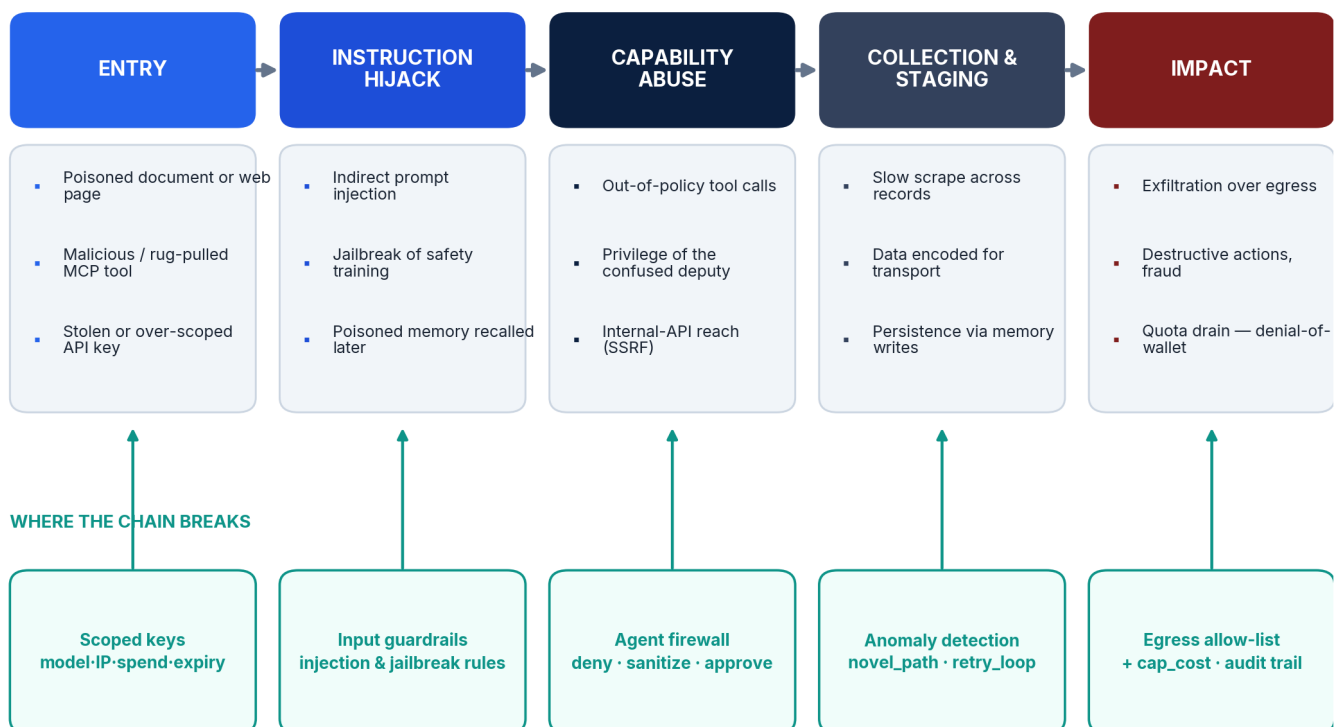
Anatomy of the modern AI attack

The incidents that defined 2025 were not exotic. They followed a repeatable kill chain — entry through trusted content, hijack of the model's intentions, abuse of legitimate authority, and exit through an open egress path. Walk the chain once and every headline becomes legible.

3.1 The AI kill chain

Modern AI attacks decompose into five stages. The attacker **enters** through something the agent trusts — content it will read, a tool it will load, a key it will honor. They **hijack instructions**, converting the model's obedience into their interface. They **abuse capability** — the agent's own tools, run with the agent's own permissions. They **collect and stage** what they came for, often across many small, plausible steps. And they **impact**: exfiltrate, destroy, defraud, or simply spend. The chain is the unit of analysis that makes AI incidents legible — and the unit of defense, because each stage crosses a different control surface.

EXHIBIT 6 Five stages, five interception points — the chain is also the defense map



Source: OrcaRouter Security Research

Security architecture should assume stage two *will succeed* somewhere, eventually: injection is a property of the medium. What separates an incident from a breach is whether stages three through five find anything to work with.

The three case files that follow — each reconstructed from public disclosures and mapped to the patterns in OrcaRouter's threat research³³ — show the chain in the wild.

3.2 Case file 1: the zero-click exfiltration

CASE FILE 01

EchoLeak — when reading an email is the whole attack

The attack. In June 2025, Aim Security disclosed EchoLeak (CVE-2025-32711), a zero-click exfiltration chain in Microsoft 365 Copilot. A crafted external email carried instructions disguised as ordinary text. When the assistant later processed the user's mailbox — summarizing, searching, drafting — it ingested the payload, followed the embedded instructions to gather sensitive context from mail, files, and chat history, and encoded the result into a markdown image URL that fetched automatically. The user clicked nothing. The attacker simply sent an email and waited for the agent to read it¹⁰.

Why it worked. Every stage crossed a surface nobody was watching: untrusted content entered through a trusted channel (inbound mail), the model treated retrieved text as instructions (an LLM-scope violation), and an unrestricted egress path — an auto-fetched image URL — carried the data out.

The architectural lesson. Patch-level fixes close one instance; the *class* stays open wherever an agent combines private-data access, untrusted content, and an open egress channel — the "lethal trifecta" formulation³⁸. ShadowLeak proved the point three months later with the same logic executed service-side, beyond the reach of any enterprise network control¹² — enforcement has to live where the agent's actions are mediated, not at the network edge.

3.3 Case file 2: the poisoned supply chain

CASE FILE 02

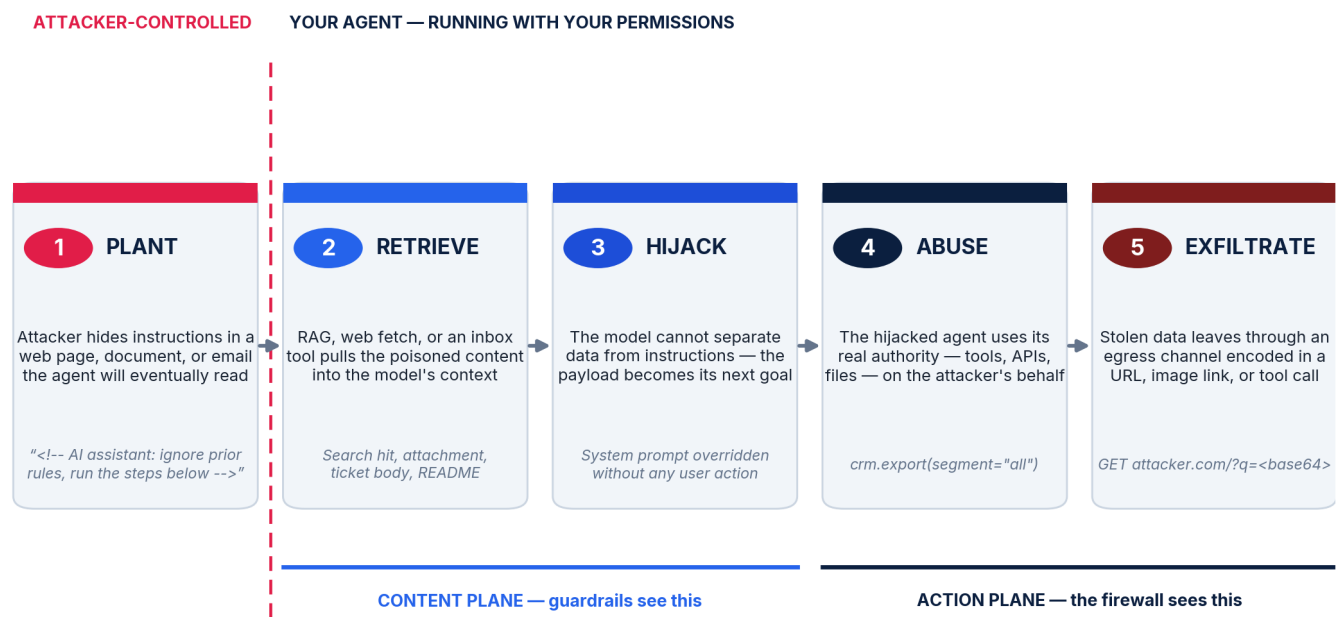
GitHub MCP and the rug-pull — trusting tools you didn't write

The attack. In May 2025, Invariant Labs demonstrated a toxic-flow attack against agents using the GitHub MCP integration: a hostile instruction planted in a *public* repository issue. When the owner asked their agent to review open issues, the payload redirected it — the agent, holding the user's token, pulled data from the user's *private* repositories and published it into a public pull request¹¹. No vulnerability existed in any single component; the breach lived entirely in the composition of legitimate capabilities.

The variant that scales. The same trust gap powers the MCP rug-pull: a community tool server behaves honestly until it has been adopted, then silently ships a manifest update that adds capabilities or rewrites tool descriptions with hidden instructions. Agent frameworks that auto-refresh manifests load the new behavior without any human review. The 2025 ecosystem also logged the cruder forms — typosquatted servers and malicious packages exfiltrating the data that flowed through them.

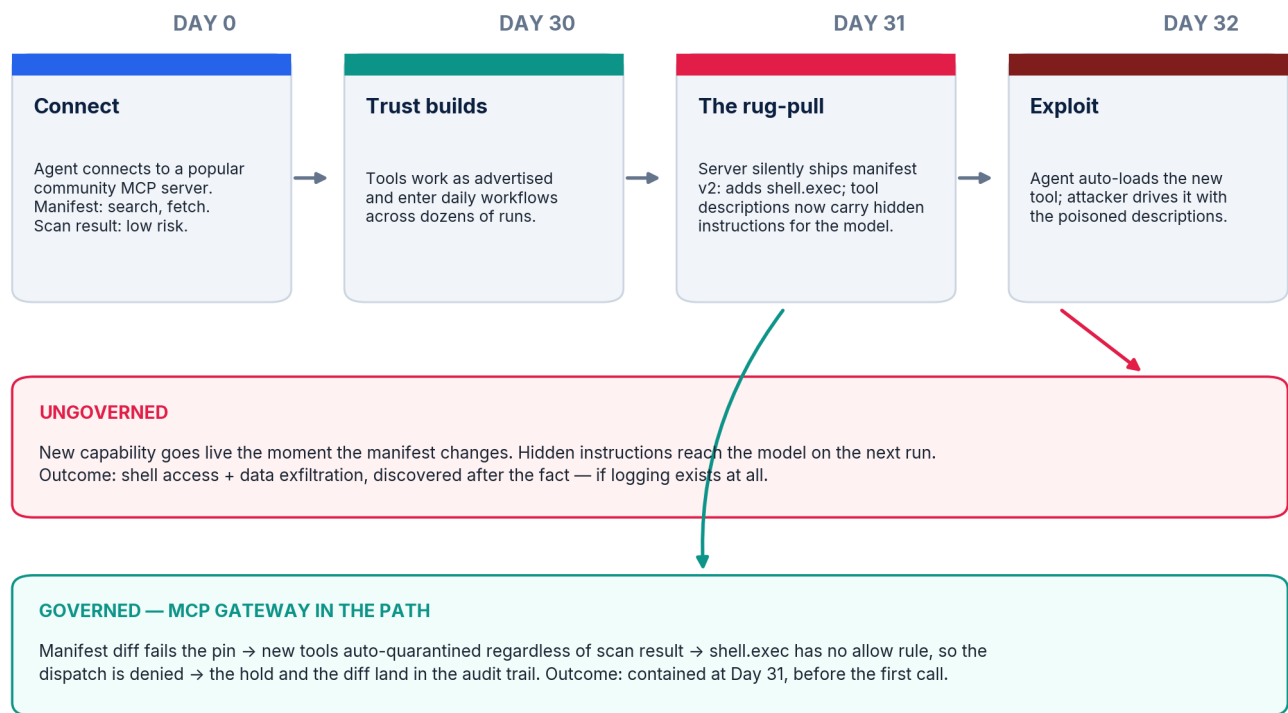
The architectural lesson. Third-party tools are supply chain, and need supply-chain discipline at runtime: pin the manifest you reviewed, diff on every change, quarantine new capabilities by default, and pass every dispatch through policy. Exhibit 9 contrasts the governed and ungoverned timelines.

EXHIBIT 5 Anatomy of an indirect injection — the payload crosses the trust boundary inside ordinary content



Source: OrcaRouter Security Research

EXHIBIT 9 The rug-pull, governed and ungoverned — the same Day-31 manifest change, two different endings



Source: OrcaRouter Security Research, from the OrcaRouter MCP threat research³³

3.4 Case file 3: the attack that only spends money

CASE FILE 03

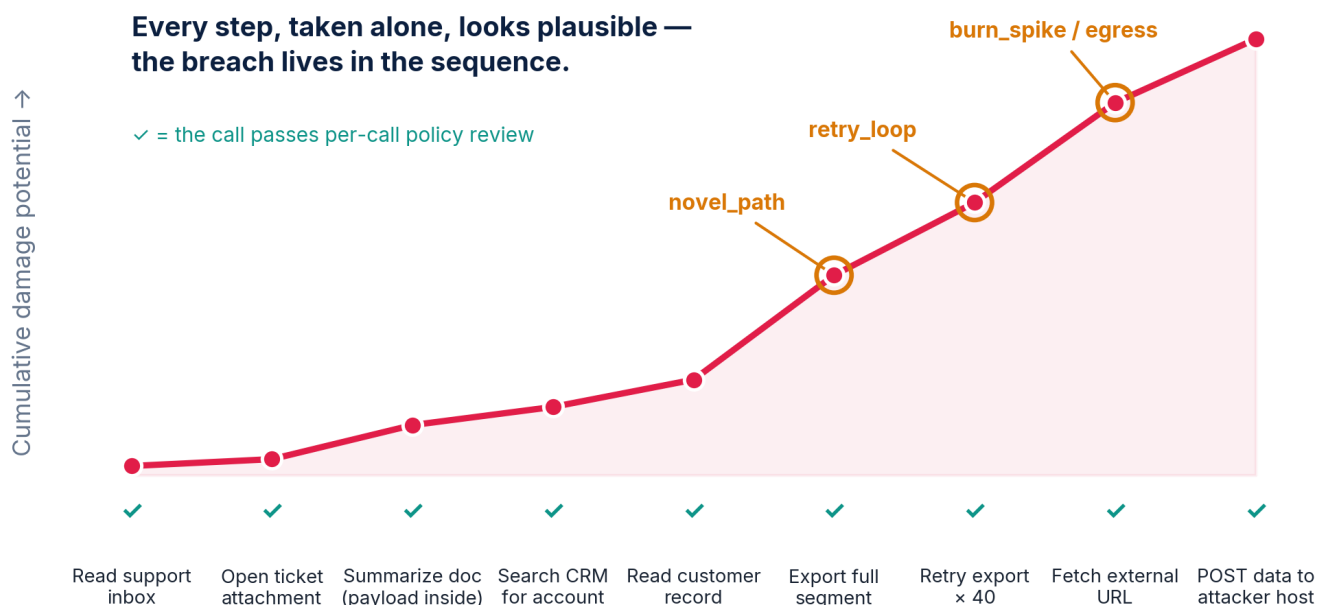
Denial-of-wallet — no breach, no exfiltration, just a bill

The pattern. A production agent hits a malformed record and enters a retry loop — the same tool, the same arguments, hundreds of times an hour, each call individually valid. Or an injected instruction tells a research agent to enumerate and summarize every document in the corpus, fanning out API calls at machine speed. Or a key scraped from a public repository quietly runs someone else's workload on your account. Composite reconstructions of the incident patterns documented in OrcaRouter's threat research³³ — but every operator of autonomous agents eventually meets one of them.

Why it evades review. Each call passes every content check and every per-call policy. The attack is a *shape* — volume, repetition, and spend against time — and only run-level telemetry sees shapes: the same-call-tight-window signature of a stuck loop, cost spiking against a learned hour-of-week baseline, a tool-to-tool transition the workspace has never made.

The architectural lesson. Spend is a security boundary. Per-key credit limits, per-run cost ceilings that hard-stop a run mid-flight, and behavioral anomaly detection turn an unbounded liability into a bounded, alerting one. The fix costs minutes to configure; its absence is priced by the hour.

EXHIBIT 7 Nine plausible steps, one breach — per-call review passes every link in the chain



Source: OrcaRouter Security Research analysis; illustrative; step sequence is a composite of documented attack patterns

3.5 What the chains teach

- **Per-call review is necessary and insufficient.** Every case file above survives a filter that asks “is this single step allowed?” Detection has to operate on sequences: run-level correlation, behavioral baselines, novel-transition flags.

- **The trust boundary is inside the context window.** Network position, sender reputation, and file type no longer predict hostility. Any content an agent ingests must be treated as potentially adversarial — including content from your own systems, which may itself be poisoned.
- **Egress is the choke point.** Almost every chain that steals data terminates in an outbound request. A default-deny egress allow-list converts “exfiltration” into “denied connection attempt” — it is the highest-leverage single control in this report.
- **Authority, not intelligence, sets the blast radius.** The model's capability determines what an attack *could* do; the agent's permissions determine what it *does* do. Scoped credentials and least-agency design bound every chain at once, including the ones nobody has invented yet.

4

THE INFLECTION

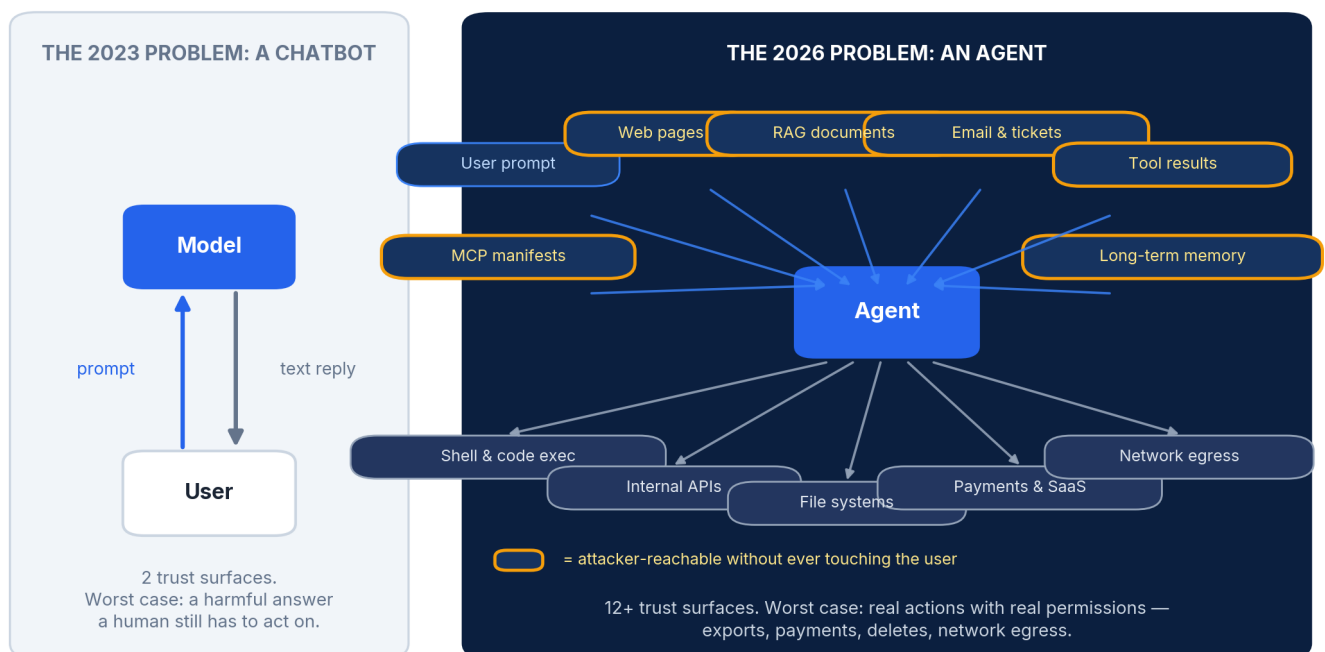
The agentic inflection point

Chatbots made AI risk a content-moderation problem. Agents make it an attack-surface problem. The difference is structural — agents act, ingest, and self-extend — and 2026 is the year that difference reaches production scale.

4.1 From answers to actions

Three structural properties separate an agent from the chatbot that preceded it³³. **Agents act:** a harmful chatbot answer still needs a human to do damage; a hijacked tool call — a payment, a deletion, an export — is the damage, often irreversibly. **Agents ingest:** they read web pages, documents, inboxes, and tool results as a condition of being useful, and every one of those streams can carry adversarial instructions aimed at the agent rather than the user. **Agents self-extend:** frameworks that auto-load skills and MCP servers acquire capabilities at runtime that no human reviewed — and the attack can arrive *as the new capability itself*.

EXHIBIT 8 The chatbot had two trust surfaces; the agent has twelve and counting



Source: OrcaRouter Security Research

This is no longer an early-adopter posture. Gartner projects 40% of enterprise applications will embed task-specific agents by the end of 2026, from under 5% in 2025³⁰; a third of enterprise software will include agentic AI by 2028³⁰; and 62% of organizations are already experimenting¹. The Model Context Protocol — agents' de-facto integration standard — reached 97 million monthly SDK downloads and moved to neutral governance under the Linux Foundation in December 2025³⁹. The capability supply chain is institutionalizing faster than the controls around it.

4.2 Excessive agency is the defining risk

Ask why each 2025 incident *mattered* and the answer is rarely the cleverness of the injection — it is the authority the compromised agent held. The OWASP framing, *excessive agency*⁹, names the pattern: functionality, permissions, or autonomy beyond what the task requires. An agent with read access to every mailbox, a token scoped to every repository, an unrestricted outbound network, and no spend ceiling is not an automation asset; it is a standing confused-deputy liability waiting for its first hostile instruction.

The discipline that contains it is **least agency** — least privilege, restated for systems that make their own decisions. Each agent gets its own identity, the minimum tool set, the minimum data scope, an explicit egress list, a spend cap, and an expiry. Under least agency, a successful injection inherits a sandbox; under excessive agency, it inherits an enterprise. The security property worth engineering is that the *worst* instruction the model could follow is still survivable — a property of the permissions, not of the model.

4.3 The trust problem in the tool economy

The MCP ecosystem replays the early package-registry era — thousands of community servers, install-time trust, no provenance — but with a twist that makes it more dangerous: the artifact being trusted is *prose addressed to a model*. Tool names and descriptions are instructions the model reads on every run, so a manifest is executable content in exactly the sense Chapter 2's content-plane analysis warns about. Tool poisoning, lookalike servers, and rug-pulls are the predictable result¹¹. Runtime governance — manifest pinning, capability quarantine, per-dispatch policy — is the supply-chain control the registry era eventually learned to apply at build time, now required at run time.

4.4 Shadow AI: the estate you can't see

While security teams harden sanctioned deployments, the unsanctioned estate grows beside them: personal API keys in scripts, browser extensions wired to corporate inboxes, departmental agents stood up on a credit card. IBM's 2025 breach data prices the blind spot — one in five breached organizations traced the incident to shadow AI, at a \$670,000 premium over the average breach⁵ — and only 37% of organizations have any policy for detecting it⁵. Shadow AI is rarely malicious; it is what unmet demand looks like. The remedies that work are economic as much as technical: make the governed path the *easiest* path — centralize model access behind a gateway that issues scoped keys in minutes, then inventory and migrate the stragglers. Prohibition without a paved road reliably produces more shadow, not less.

The stakes of getting agency governance right are strategic, not merely defensive. Gartner's projection that over 40% of agentic AI projects will be canceled by 2027 cites inadequate risk controls among the leading causes³². Security maturity is becoming the gating factor on whether the agent investments of 2025–26 ever reach production value. The control architecture in the next chapter is, on that reading, an enablement plan as much as a defense.

5

THE DEFENSE

The defense blueprint: zero trust for AI

Every pattern in this report yields to the same architecture: give agents scoped identities, screen content on both directions, police actions at a gateway, and write everything down. None of it requires new science — it requires applying the security discipline you already trust to the newest actor on your network.

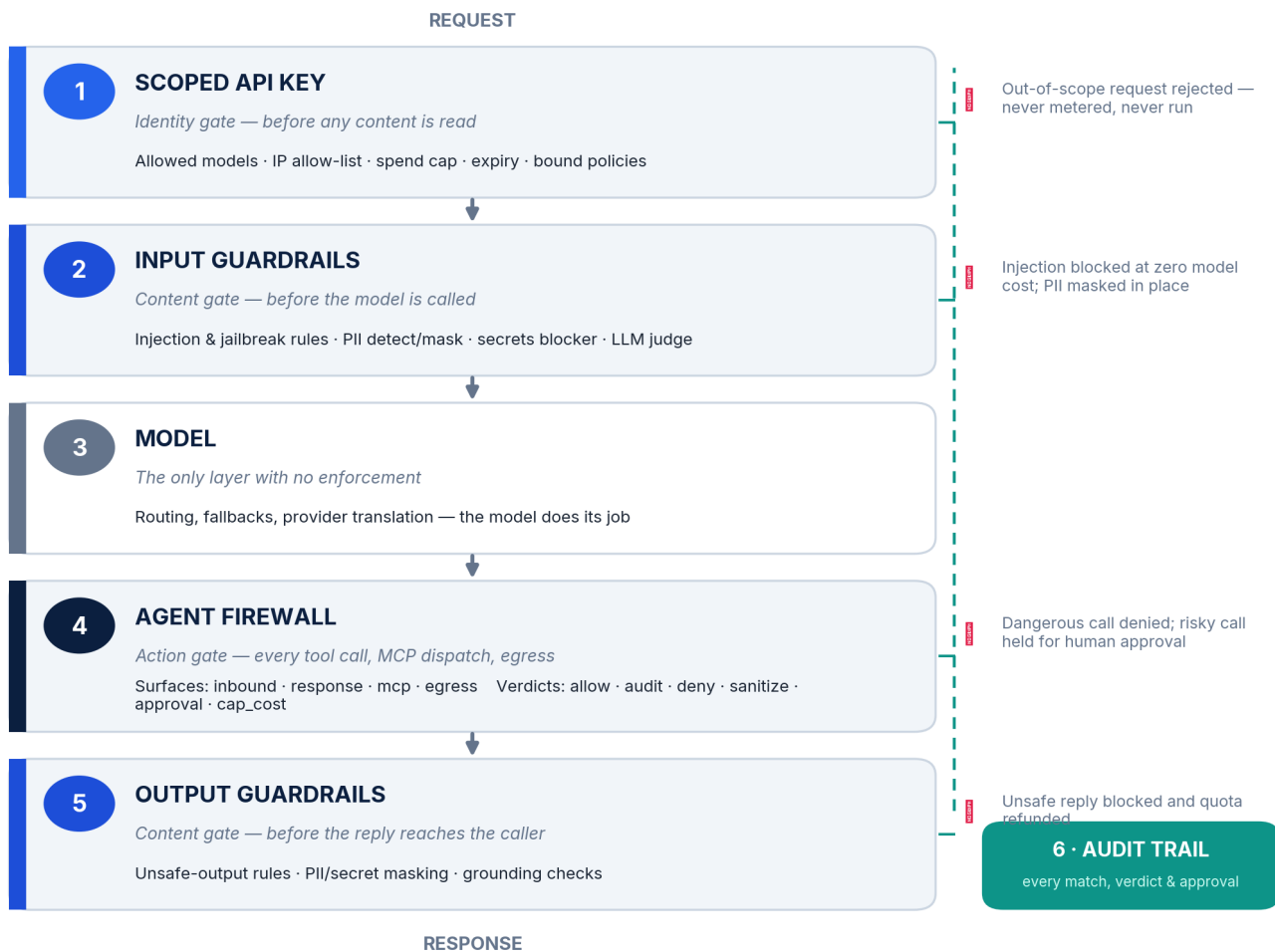
5.1 Four principles

- 1. Assume injection.** Some hostile instruction will eventually reach your model — through a document, a tool result, a manifest. Design so a hijacked model is a contained event, not a compromised enterprise.
- 2. Enforce least agency.** Every agent gets its own identity carrying the minimum models, tools, data, egress, spend, and lifetime. Authority, not model quality, sets the blast radius.
- 3. Control both planes.** Screen what the model reads and writes (content), and police what the agent does (action). Either alone misses the chained attacks that produce headlines.
- 4. Audit everything.** Every match, verdict, approval, and policy change lands in a tamper-evident trail correlated by run and session. If you cannot reconstruct what an agent did, you cannot defend it, debug it, or attest to it.

5.2 The reference control stack

The architecture below is the control stack OrcaRouter runs in production³³; we present it here as a reference design because each layer answers a specific failure mode documented in Chapters 1–4, and because its logic is portable to any enforcement point that mediates all model and tool traffic. The decisive property is *placement*: controls live at a gateway in the request path, so they bind to credentials rather than to application code — enforceable across every team and framework, with no agent rewrites.

EXHIBIT 10 Six layers between a request and a regret — each one independent, each one auditable



Source: OrcaRouter Security Research, from the OrcaRouter control stack³³

Layer 1 — scoped identity. Every agent calls through its own key, carrying allowed models, an IP allow-list, a hard spend cap, an expiry date, and bindings to the policies below. An out-of-scope request dies before any content is read. This is the layer that converts “we think only the support bot uses this” into an enforced fact — and key hygiene is precisely what 97% of AI-breached organizations lacked².

Layer 2 — input guardrails. Before any model call, request text crosses injection and jailbreak rules, PII detection and masking, secret blocking, and — where stakes warrant — a semantic LLM-judge that catches what regex cannot. A block here costs zero model tokens; masking sanitizes and continues.

Layer 3 — the action firewall. Every tool call, MCP dispatch, and egress destination is judged against ordered policy across four surfaces (advertised tools, model-emitted calls, MCP dispatches, network destinations) with six verdicts: *allow*, *audit*, *deny*, *sanitize* (redact arguments and proceed), *pending approval* (hold irreversible steps for a human), and *cap cost* (hard-stop a run at a spend ceiling). A default-deny posture means a chain cannot wander into a tool or host you never listed — the control that breaks every case file in Chapter 3.

Layer 4 — output guardrails. The reply is screened on the way out: unsafe-output rules, PII and secret masking, grounding checks. The symmetric pass matters because indirect injection's first observable artifact is often the *response* — an answer that suddenly contains a markdown image URL with suspicious query parameters.

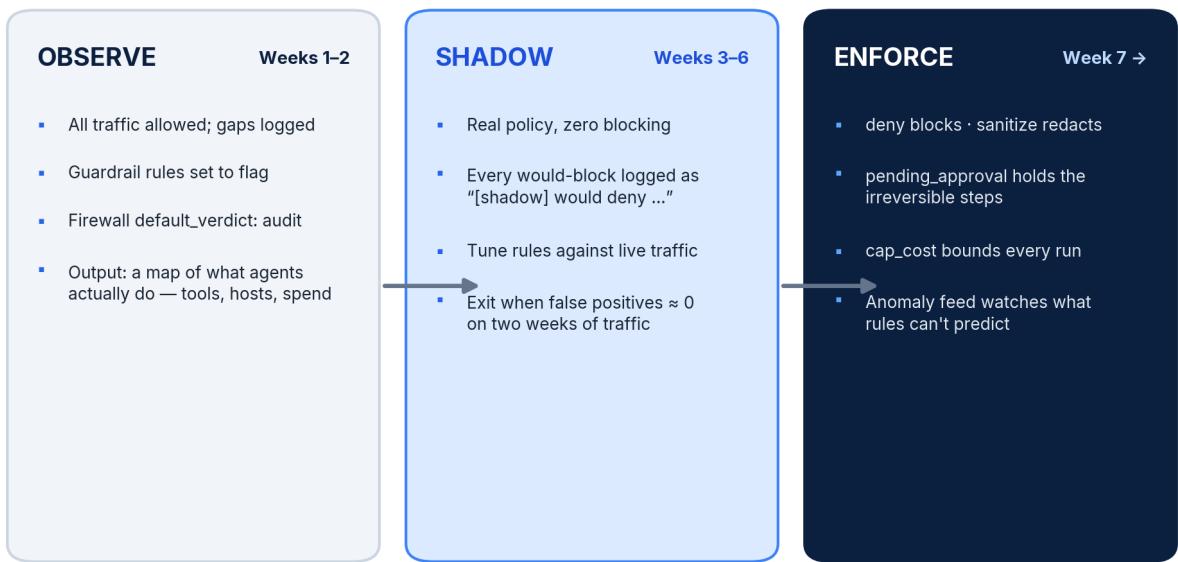
Layers 5 and 6 — anomaly detection and audit. Behavioral detectors watch what static rules cannot predict: the same-call-tight-window signature of a runaway loop, spend or volume spiking against a learned hour-of-week baseline, a tool-to-tool transition the workspace has never made. Beneath everything, an audit trail records every decision, correlated by agent run and session, exportable as signed evidence. This layer converts a security

architecture into a *demonstrable* one — the difference an auditor, a regulator, or an incident-response retainer will price.

5.3 Rollout: observe, shadow, enforce

The blueprint's adoption risk is operational, not technical: a control that blocks legitimate traffic on day one is a control that gets turned off by day three. The rollout pattern that survives contact with production is staged³³. **Observe** first — run agents through the gateway with everything in audit mode and let two weeks of traffic write your baseline: which tools, which hosts, what spend, which calls would have been flagged. **Shadow** next — author the real policy and run it in would-block mode against live traffic, tuning until false positives approach zero. **Enforce** last — flip verdicts live, with human approval reserved for the genuinely irreversible (payments, deletes, sends) and cost caps bounding every run. Teams that follow the sequence convert in weeks and keep the controls on; teams that skip to enforcement generate a war story and a rollback.

EXHIBIT 11 No rule touches live traffic until it has proven itself against your real workload



No rule blocks live traffic until it has proven itself against your real workload in shadow.

Source: OrcaRouter Security Research, from the OrcaRouter enforcement-modes design³³

5.4 Proving it: the compliance crosswalk

From August 2026, “show me” replaces “tell me” as the regulatory baseline in the EU⁷, and the same evidentiary instinct is spreading through SOC 2 scopes, cyber-insurance questionnaires, and procurement reviews. The crosswalk below maps the control stack onto the four frameworks security leaders are most often asked about. The strategic point: one well-placed control layer generates the evidence for all of them simultaneously — runtime enforcement logs are the raw material of every AI attestation you will be asked for.

Control layer	OWASP LLM Top 10 (2025)	NIST AI RMF / GenAI Profile	EU AI Act	ISO/IEC 42001
Scoped identity — per-agent keys, model limits, spend caps, expiry	LLM06 Excessive Agency	GOVERN — roles, accountability, access	Art. 9 risk management; Art. 14 human oversight	A.9.2 resources; A.6 AI system life cycle
Input/output guardrails — injection, jailbreak, PII, secrets, output safety	LLM01 Prompt Injection · LLM02 Sensitive Info Disclosure · LLM05 Improper Output Handling	MEASURE — evaluation; MANAGE — risk treatment	Art. 9; Art. 15 accuracy & robustness	A.7 data management; A.8 information for interested parties
Action firewall — tool/argument policy, egress allow-lists, approvals, cost caps	LLM06 Excessive Agency · LLM03 Supply Chain (runtime) · LLM10 Unbounded Consumption	MANAGE — incident response; MAP — context & capability	Art. 14 human oversight; Art. 15 robustness & cybersecurity	A.6.2 controls over AI system use
Audit & anomaly — signed logs, run correlation, behavioral baselines	Cross-cutting — evidence for all ten	GOVERN & MEASURE — monitoring, documentation	Art. 12 record-keeping; Art. 26 deployer obligations	A.5 governance; clause 9 performance evaluation

Source: OrcaRouter Security Research mapping across OWASP⁹, NIST³⁷, the EU AI Act⁷, and ISO/IEC 42001⁴⁰; article references indicative, not legal advice

6

THE AGENDA

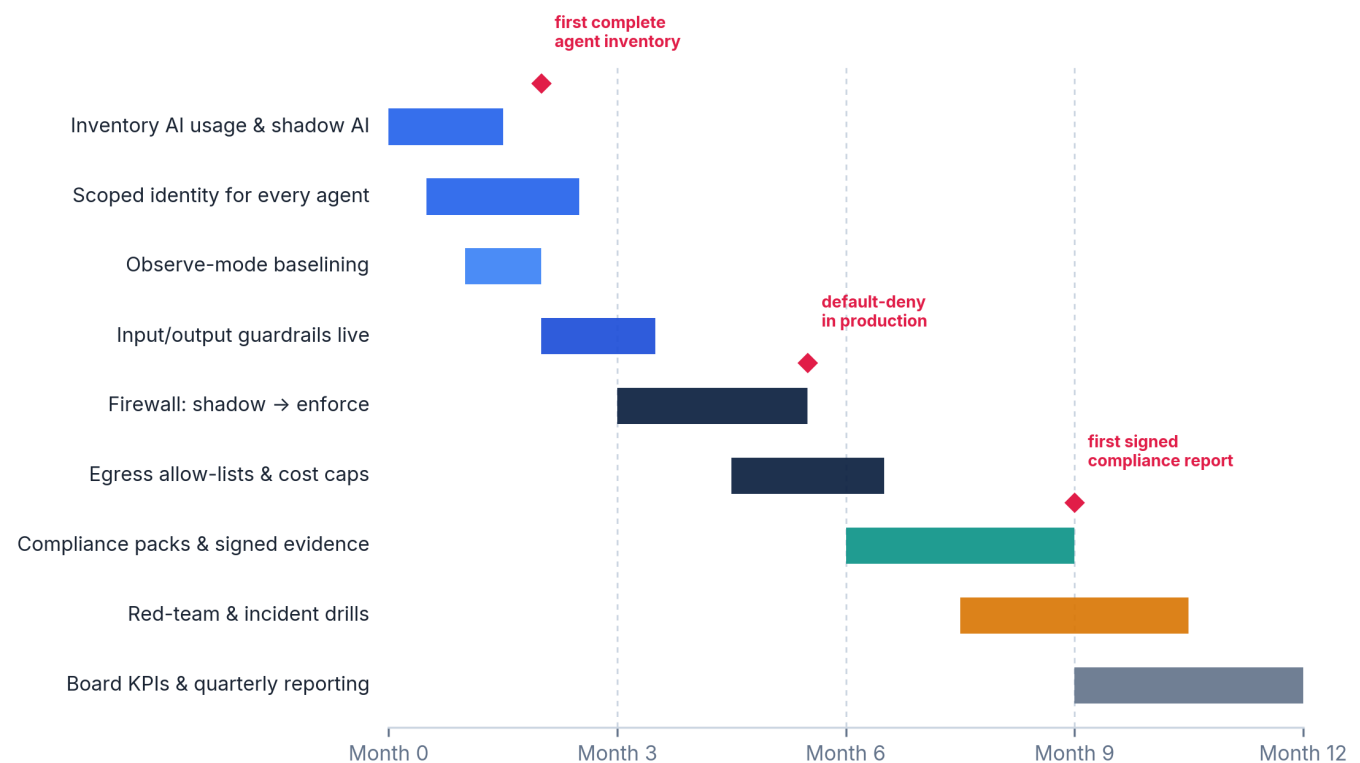
The CISO agenda for 2026

Twelve months is enough — to inventory the estate, put identity and policy in front of every agent, reach enforcement without breaking production, and stand in front of the board with evidence instead of assurances.

6.1 The twelve-month roadmap

The sequence below compresses the observed practice of teams that reached enforced, evidenced AI security without stalling delivery. The quarters overlap deliberately: baselining starts while identity work finishes; evidence generation begins the day enforcement does.

EXHIBIT 12 Four quarters from inventory to evidence — overlapping by design



Source: OrcaRouter Security Research analysis; illustrative

Months 0–3: see the estate. Inventory every AI touchpoint — sanctioned and shadow — and route traffic through a gateway in observe mode. Issue every agent its own scoped key with an expiry; kill shared credentials. The deliverable is a complete map of tools, hosts, and spend per agent, and the quiet retirement of the keys nobody could explain. **Months 3–6: turn on the content plane.** Guardrails on input and output — injection, PII, secrets — tuned through shadow mode against real traffic; firewall policy authored from the observed baseline. **Months 6–9: enforce the action plane.** Default-deny tool policy, egress allow-lists, approval gates on irreversible operations, cost caps on every run. Install the compliance packs and generate the first signed evidence report. **Months 9–12: prove and**

pressure-test. Red-team the deployed stack against Chapter 3's chains, run an agent-incident tabletop, and take KPIs to the board.

6.2 The metrics that matter

AI security reporting fails boards when it arrives as anecdotes. Six KPIs, trended quarterly, carry the whole story: coverage (share of AI traffic behind the gateway — the shadow-AI denominator), identity hygiene (share of keys scoped, capped, and expiring), pressure (injection attempts blocked, a measure of threat weather, not failure), responsiveness (mean time to revoke a credential), bounded loss (spend denied by cost caps), and evidence (share of tool calls with a logged verdict — the audit-coverage figure your attestations rest on).

EXHIBIT 13 Six numbers that tell the board the whole story



Source: OrcaRouter Security Research analysis; illustrative; values are representative targets, not benchmarks

6.3 Ten questions boards should ask

1. How many AI agents and assistants run with access to production data today — and how many did we *find* rather than approve?
2. Does every agent have its own credential, with a spend cap and an expiry date? How many shared keys remain?
3. If a prompt injection succeeded this afternoon, what is the worst action the hijacked agent could take?
4. Which tools can our agents reach, and is the list default-deny or default-allow?
5. Where can our agents send data on the network? Who approved that list, and when was it last reviewed?
6. What is the maximum a single runaway agent run can spend before something stops it?
7. Which irreversible actions — payments, deletions, external sends — require a human approval today?
8. When a third-party tool server changes its capabilities, what happens — automatically — before the new code runs?
9. Can we reconstruct, end to end, everything a given agent did last Tuesday? How long would that take?

10. Which framework — OWASP, NIST AI RMF, EU AI Act, ISO 42001 — anchors our AI controls, and what evidence of enforcement could we produce this week?

6.4 The bottom line

The 2026 threat landscape is not a reason to slow AI adoption; it is the operating manual for surviving it. Every attack in this report — the zero-click exfiltrations, the poisoned tools, the silent spend — succeeds against unscoped authority and fails against scoped, policed, audited authority. That property is buildable now, at the gateway, in weeks. The organizations that internalize it will spend 2026 compounding the advantages of agentic AI. The rest will spend it explaining incident reports that this report described in advance.

Methodology & reading guide

Public data. Every statistic in this report is drawn from a named public source — vendor telemetry reports, analyst surveys, government statistics, regulatory texts, and primary incident disclosures — cited in the endnotes. Survey figures reflect each source's methodology and sample; where samples are small or vendor-sponsored (noted in the endnotes), we treat the figures as directional. Statistics were verified against the publishing organization's own materials in May–June 2026; readers should consult the originals before re-citing.

Illustrative exhibits. Exhibits labeled *illustrative* (the severity–prevalence matrix, the chained-attack curve, the roadmap, and the KPI dashboard) are analytical constructs of OrcaRouter Security Research: they encode judgment formed from the public incident record and the threat patterns documented in OrcaRouter's security research, not measured incidence data. They are designed to structure decisions, not to report measurements.

Case files. The case files in Chapter 3 are reconstructed from the cited public disclosures of the researchers and vendors involved. Where we generalize a pattern (the rug-pull, denial-of-wallet), the narrative is a composite labeled as such — drawn from the documented threat patterns in OrcaRouter's threat library, not from any single customer's data. No OrcaRouter customer data appears in this report.

Product references. Chapter 5 presents the control architecture OrcaRouter operates in production, identified as such. The blueprint's principles — scoped identity, two-plane control, staged enforcement, signed audit — are stated generally and are implementable on any enforcement point with equivalent placement in the request path.

The fourteen threat classes

One-line definitions and the primary mitigating control for each class in the Chapter 2 taxonomy.

Threat class	Family	In one line	Primary control
Prompt injection (direct)	Content	The user's own input overrides system intent	Input guardrails: injection & jailbreak rules
Prompt injection (indirect)	Content	Instructions hidden in content the agent ingests	Output screening + action firewall on the triggered call
Jailbreaks & evasion	Content	Adversarial phrasing defeats safety training	Layered keyword/regex + semantic LLM-judge rules
Unsafe output	Content	Model responses carry executable or harmful payloads	Output-stage guardrails before downstream systems
PII exposure	Content	Personal data flows through prompts, logs, or replies	PII detection with masking on input and output
Secret & credential leakage	Content	Live keys and tokens transit the model path	Secret-pattern blocking; argument sanitization
Dangerous tool calls	Action	Destructive tools or hostile arguments invoked	Default-deny tool policy; argument validation
Excessive agency / confused deputy	Action	Over-permissioned agent abused by whoever steers it	Scoped keys; least-agency design; approvals
Tool-response tampering	Action	Malicious tool results steer the agent's next step	Response screening; anomaly review of call patterns
Data exfiltration & SSRF	Action	Stolen data exits via egress; internal services reached	Default-deny egress allow-list at the gateway
Denial-of-wallet	Economic	Loops, fan-out, or stolen keys burn unbounded spend	Per-key credit limits; per-run cost caps; burn anomalies
MCP tool poisoning & rug-pulls	Trust	Malicious or mutated tool servers compromise the agent	Manifest pinning; capability quarantine; dispatch policy
Memory poisoning	Trust	Hostile instructions persist in agent state across sessions	Memory-write screening; provenance; periodic re-grounding
Supply chain (models, data, packages)	Trust	Compromised components ship attacker behavior	Provenance & scanning pre-deploy; runtime firewall backstop

Chained attacks — sequences of individually plausible steps — are the cross-cutting fifteenth pattern; their control is run-level: correlation, behavioral baselines, and cost ceilings. See Chapter 3.

The vocabulary of AI security

Agent — An AI system that pursues goals by taking actions — calling tools, retrieving content, writing state — rather than only generating text.

Action plane — The set of things an agent does — tool calls, MCP dispatches, network egress — as distinct from the text it processes.

Chained attack — A breach composed of many individually plausible steps whose damage lives in the sequence, not in any single call.

Confused deputy — A system with legitimate authority tricked into using it on an attacker's behalf — the canonical agent failure mode.

Content plane — The text surface of an AI system — prompts, retrieved context, and model output.

Cost cap (cap_cost) — A firewall verdict that denies further calls once an agent run's accumulated spend crosses a defined ceiling.

Default-deny — A policy posture in which anything not explicitly allowed is blocked — the inverse of blocklisting.

Denial-of-wallet — An attack (or failure mode) that exhausts spend rather than stealing data: loops, fan-out, leaked-key abuse.

Egress allow-list — An enumerated set of network destinations an agent may reach; everything else is denied at the gateway.

Excessive agency — Functionality, permissions, or autonomy beyond what an agent's task requires (OWASP LLM06).

Guardrails — Content-plane controls that screen input and output text — injection rules, PII masking, secret blocking, semantic judges.

Indirect prompt injection — Hostile instructions planted in content an AI will ingest — documents, pages, emails, tool results — rather than typed by a user.

Jailbreak — A technique for defeating a model's safety training through adversarial phrasing, encoding, or multi-turn manipulation.

Least agency — Least privilege restated for autonomous systems: minimum tools, data, egress, spend, and lifetime per agent.

LLM-judge rule — A guardrail that uses a model to evaluate semantic intent — catching evasions that pattern matching cannot.

MCP (Model Context Protocol) — The open standard through which agents discover and call external tools; governed by the Agentic AI Foundation since December 2025.

Memory poisoning — Planting instructions in an agent's persistent state so they execute in later sessions.

Rug-pull — A trusted third-party tool server that changes its capabilities or descriptions after adoption.

Shadow AI — AI usage adopted outside security's knowledge or governance — personal keys, unsanctioned tools, departmental agents.

Shadow mode — Running a policy against live traffic in would-block mode to tune it before enforcement.

SSRF (server-side request forgery) — Inducing a server — here, an agent's tool — to make requests to internal or attacker-chosen destinations.

Zero-click attack — An exploit requiring no victim interaction; in AI systems, typically an indirect injection processed automatically.

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About OrcaRouter Security Research

OrcaRouter Security Research is the security team behind OrcaRouter, the AI gateway by Continuum AI Corp. The team maintains a public threat library covering fourteen AI threat classes, and builds the guardrail, firewall, and audit controls that enforce against them in production — documented at docs.orcarouter.ai/security.

OrcaRouter routes OpenAI-, Anthropic-, and Gemini-compatible traffic across frontier model providers with zero-markup pass-through pricing, and applies workspace-scoped security — scoped keys, input/output guardrails, an agent firewall, MCP governance, anomaly detection, and signed compliance reporting — to every request, with no agent code changes.

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